

Q Why do startups and SMEs need to focus on open source deployments instead of some closed source solutions?

I recently read the article 'Programming By Poking: Why MIT Stopped Teaching SICP (The Structure and Interpretation of Computer Programs)'. It's the course that taught many software engineers how to write good code. The course has now been abandoned because engineers have changed. Today, developers use libraries without fully understanding what happens under the hood of the library. To quote Professor Sussman: "You grab this library and you poke at it. You write programs that poke it and see what it does. And you say: 'Can I tweak it to do the thing I want?'"

I see this happening every day when I'm on StackOverflow. It's as if developers nowadays only know one book, titled: 'Copying and Pasting from StackOverflow.' I answer questions from people who say, "We don't care about how iText works, we need iText to do this!" I often wonder why developers who say such things aren't ashamed. Where is their honour?

I can understand why people use libraries. I use libraries myself because the world has gotten so complex that you can no longer master every discipline yourself and it would be stupid to reinvent the wheel. It's only normal that you try to reuse work that has been done before.

Nevertheless, you should have some knowledge of the tools you're using. You don't want your entire business to depend on software that you don't understand and that is completely closed. With open source, you always have the option of looking at how the code is written and you are free to adapt the software to your own needs. That is a huge advantage. Even if you never bother to look under the hood, you know that you can do so if you ever need to.

Q How can India play a vital role in expanding the open source world?

I am the first person in my family who went to college. My mother went to school until the age of 16, and my father until the age of 18. I know that there are many countries where going to school until the age of 16 or 18 isn't common, but in Western terms, it's unusual. As a child of people without a high-school diploma, I was a nobody and I knew that it wouldn't be trivial for me to succeed in life.

Access to the Internet and the rise of open source made it possible for me to make the difference. Who I was, where I was born, where I lived—all of that didn't matter on the Internet. I see that some Asian countries block the Internet. They don't allow their citizens to have access to resources that could help them to become successful. That's a recipe for disaster.

When I look at India, I see a huge population of well-educated people. At the Great Indian Developer Summit

(GIDS), I saw a large audience of talented developers. Moreover, the percentage of female attendees at GIDS was much higher than in most of the other developer conferences I attended in the rest of the world. All of these observations tell me that India is doing the right things.

As for open source, when I look at the number of visits to the iText site, India is ranked second after the US. When I look at our worldwide revenue, India doesn't appear in the top 15. In the past, many Indian companies just used iText without ever giving anything in return, often out of ignorance about the open source licence. This is now slowly changing; commercial open source business is growing in India. I can only speak for iText, but now that we see our business growing in India, our trust is also growing and we're more inclined to invest in India.

India can play a vital role in expanding the open source world by allowing this vast resource of talented people to contribute to open source projects. When you write closed source software, you can't show the world how talented you are. Open source software is very transparent; everyone can assess the quality of your code. Ultimately, this should improve the quality of code in general.

“The most essential factor for an open source software project (any open source project) is whether people are willing to give something back for their use of the software.”

Q What's your take on the importance of a community to make an open source solution successful in the market?

Large companies want to use software and do whatever they want with it. They don't like GPL-style licences because those limit what they can do

with the software for free. Usually, such companies are not in the business of selling software. They might be offering software as a service like Google; using a totally different business that relies on software such as Amazon; selling closed source, proprietary products that use open source software internally like Oracle and IBM; or operating their primary business on advertisements, and viewing their users as their product such as Facebook.

Companies like this promote the use of liberal licences such as the Apache Software License. When a developer publishes his source code as Apache software, these companies can use that software without ever having to pay that developer. It's usually sufficient to sponsor a foundation as long as that foundation succeeds in convincing sufficient developers to create code. This is a great way to make money with open source, but it has some disadvantages. The moment a large corporation decides that the value created by a project doesn't justify the investment, the 'charity' will stop. Oracle dropped GlassFish, IBM backed away from Geronimo, and Pivotal left Groovy.

If this trend continues, an individual, independent open source developer will disappear and open source will only